

Member Task Assignment & Reporting Guide: Guevarra, John Angel D.

Role in Project: Project Manager / Systems Analyst & Deployment Lead

Primary Chapters: Chapter 3, Chapter 4

★ Assigned Sections Checklist

Chapter 3: Technical Background

☐ 3.1 Development

- ☐ 3.1.1 Hardware (Personal Computers and Laptops used in development)
- ☐ 3.1.2 Software:
 - ☐ A. Frontend (RPG Maker MZ, HTML5/WebGL, CSS)
 - ☐ B. Backend & Scripting (JavaScript ES6, VS Code, Node.js, PeerJS)
 - ☐ C. Multimedia & Asset Tools (GIMP, Canva, Draw.io)
 - ☐ D. Deployment & Runtime Platforms (NW.js, Chrome/Edge, GitHub, Git, Vercel, Hostinger)

Chapter 4: Methodology, Results and Discussion

☐ 4.1 Prototyping Model (SDLC Execution)

- ☐ 4.1.1 Requirements Gathering (Curriculum analysis, Prodigy benchmarking)
- ☐ 4.1.3 Building Prototype (Core engine integration, plugin development, biome maps)
- ☐ 4.1.4 Customer Evaluation of Prototype (Beta evaluation with Grade 4–6 learners)
- ☐ 4.1.5 Refining Prototype (Bug fixes, gameplay curve balancing, keypad optimization)
- ☐ 4.1.6 Engineer Product (NW.js desktop executable compilation & Hostinger web deployment)

☐ 4.2 Requirements Specification

- ☐ 4.2.1 Operational Feasibility:
 - ☐ Fishbone Diagram (Ishikawa Root Cause Diagram Analysis)
- ☐ 4.2.2 Technical Feasibility:
 - ☐ 4.2.2.1 Compatibility Checking (Hardware & Software Compatibility)
 - ☐ 4.2.2.2 Relevance of the Technology (Why JS ES6, PeerJS, NW.js, WebGL are superior for this study)
- ☐ 4.2.3 Schedule Feasibility:
 - ☐ Gantt Chart Schedules (Tables #1 to #11: Nov 2025 to Sept 2026 timelines)
- ☐ 4.2.4 Economic Feasibility:
 - ☐ 4.2.4.1 Cost and Benefit Analysis (CBA Table 13: Software licenses, Web hosting, Domain — Total ₱13,733.60)
 - ☐ 4.2.4.2 Cost Recovery Scheme (Itch.io Pay-What-You-Want model & optional DLC Infinite Dungeon expansion)

☐ 4.4 Development Specifications

- ☐ 4.4.1 Software Specification (Table 14: Complete development software inventory)

☐ 4.8 Installation Processes

- ☐ Steps to Access the Web Application (5-step browser guide)
- ☐ Steps to Download and Install the Desktop Application (7-step ZIP extraction & NW.js launch guide)

In Chapter 3, you have to report:

1 Development Tools & Software Stack (3.1.1 & 3.1.2):

- Explain the full technical stack used by the team.
- Justify why **RPG Maker MZ** was chosen as the 2D foundation (it provides rapid grid mapping, database management, and native JavaScript plugin hooks).
- Detail the role of **NW.js** (Node-Webkit) as the native Windows wrapper that allows the HTML5/JS web game to compile into an offline `.exe`.
- Explain how **GitHub/Git** managed version control, and how **Vercel** was used for rapid staging before final production deployment on **Hostinger**.

In Chapter 4, you have to report:

1 The Prototyping Model SDLC (4.1):

- Walk through the 6 iterative phases (Requirements Gathering $\rightarrow$ Quick Design $\rightarrow$ Building Prototype $\rightarrow$ Customer Evaluation $\rightarrow$ Refining Prototype $\rightarrow$ Engineer Product).
- Defend why the Prototyping Model was chosen over Waterfall: game balance, timer speeds, and math pacing cannot be fixed upfront; they require real-time iterative testing with children.

2 Operational Feasibility & Root Cause (4.2.1):

- Present the **Fishbone Diagram (Ishikawa)** to explain the root causes of math anxiety and poor drill engagement (Environment, People, Technology, Process factors).

3 Technical & Schedule Feasibility (4.2.2 & 4.2.3):

- Defend the technological relevance of JavaScript and WebGL.
- Present the **Gantt Charts (Nov 2025 to Sept 2026)** to prove project milestones, adviser consultations, revisions, and defense dates were systematically met.

4 Economic Feasibility & Cost Recovery (4.2.4):

- Present the **Cost and Benefit Analysis (CBA)** showing the total development cost of **₱13,733.60** (RPG Maker MZ licenses, Hostinger 4-year hosting/domain).
- Explain the **Cost Recovery Scheme**: Chronicles of Arithmos is 100% free for educational use, but costs are sustained through optional donations on **Itch.io** and an optional Infinite Dungeon DLC.

5 Software Specification & Installation (4.4.1 & 4.8):

- Walk through the Software Specification table (Table 14).
- Clearly guide the panel on how users access the live web link (<https://chroniclesofarithmos.site>) and how Windows users extract and execute the standalone PC build.